

AUTOMATIC POSITION REPORTING SYSTEM

The APRS Working Group

29 August 2000

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AUTOMATIC POSITION REPORTING SYSTEM

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Cover Three

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Prelude

Preface: Editor's Note

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FOREWORD

This APRS Protocol Reference document represents the coming-of-age of WB4APR's baby. Starting with a simple concept — a way to track the location of moving objects via packet radio — programs using the APRS protocol have grown into perhaps the most popular packet radio application in use today. It's also become one of the most complex; from the simple idea grew, and still grows, a tactical communications system of tremendous capability. Like many ham projects, the APRS protocol was designed as it was being implemented, and many of its intricacies have never been documented.

Until now. This specification defines the APRS on-air protocol with a precision and clarity that make it a model for future efforts. The work done by members of the APRS Working Group, as well as Technical Editor Ian Wade, G3NRW, should be recognized as a tremendous contribution to the packet radio art. With this document available, there is now no excuse for any developer to improperly implement the APRS protocol.

As an APRS Working Group member whose role was mainly that of observer, I was fascinated with the interplay among the APRS authors and the Technical Editor as the specification took form. Putting onto paper details that previously existed only in the minds of the authors exposed ambiguities, unconsidered consequences, and even errors in what the authors thought they knew. The discussion that followed each draft, and the questions Ian posed as he tried to wring out the uncertainties, gave everyone a better understanding of the protocol. I am sure that this process has already contributed to better interoperability among existing APRS applications. Everyone who has watched the specification develop, from the initial mention in April 1999 until release of this Version 1.0 document in August 2000, knows that the process took much longer than was hoped. At the same time, they saw the draft transformed from a skeleton into a hefty book of over 110 pages. With the specification now in hand, I think we can all say the wait was worth it. Congratulations to the APRS Working Group and, in particular, to G3NRW, for a major contribution to the literature of packet radio.

John Ackermann, N8UR
TAPR Vice President and APRS Working Group Administrative Chair
August 2000

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Preamble

APRS Working Group

The APRS Working Group is an unincorporated association whose members undertake to further the use and enhance the value of the APRS protocols by (a) publishing and maintaining a formal APRS Protocol Specification; (b) publishing validation tests and other tools to enable compliance with the Specification; (c) supporting an APRS Certification program; and (d) generally working to improve the capabilities of APRS within the amateur radio community.

Although the Working Group may receive support from TAPR and other organizations, it is an independent body and is not affiliated with any organization. The Group has no budget, collects no dues, and owns no assets. The current members of the APRS Working Group are:

- John Ackermann, N8UR Administrative Chair & TAPR Representative
- Bob Bruninga, WB4APR Technical Chair, founder of APRS
- Brent Hildebrand, KH2Z Author of APRS+SA
- Stan Horzempa, WA1LOU Secretary
- Mike Musick, N0QBF Author of pocketAPRS
- Keith Sproul, WU2Z Co-Author of WinAPRS/MacAPRS/X-APRS
- Mark Sproul, KB2ICI Co-Author of WinAPRS/MacAPRS/X-APRS

Acknowledgements

This document is the result of contributions from many people. It includes much of the material produced by individual members of the Working Group.

In addition, the paper on the Mic-E data format by Alan Crosswell, N2YGK, and Ron Parsons, W5RKN was a useful starting point for explaining the complications of this format.

Document Version Number

Except for the very first public draft release of the APRS Protocol Reference, the document version number is a 3-part number “P.p.D” (for an approved document release) or a 4-part number “P.p.Dd” (for a draft release):

Thus, for example:

- Document version number “1.2.3” refers to document release 3 covering APRS Protocol Version 1.2.
- Document version number “1.2.3c” is draft “c” of that document.

Release History

The release history for this document is listed in Appendix 7.

Document Conventions

This document uses the following conventions:

- Courier font ASCII characters in APRS data.
- `␣` ASCII space character.

- ... (ellipsis) zero or more characters.
- /\$ Symbol from Primary Symbol Table.
- \\$ Symbol from Alternate Symbol Table.
- 0x hexadecimal (e.g. 0x1d).
- All callsigns are assumed to have SSID -0 unless otherwise specified.
- **Yellow marker** (appears as light gray background in hard copy). Marks text of interest — especially useful for highlighting single literal ASCII characters (e.g. ") where they appear in APRS data ¹.
- Shaded areas in packet format diagrams are optional fields.

Feedback

Please address your feedback or other comments regarding this document to the TAPR aprsspec mail list.

To join the list, start at <http://www.tapr.org> and then follow the path Special Interest Groups ⇒ APRS Specification ⇒ Join APRS Spec Discussion List.

Authors' Foreword

This reference document describes what is known as APRS Protocol Version 1.0, and is essentially a description of how APRS operates today. It is intended primarily for the programmer who wishes to develop APRS compliant applications, but will also be of interest to the ordinary user who wants to know more about what goes on “under the hood”. It is not intended, however, to be a dry-as-dust, pedantic, RFC-style programming specification, to be read and understood only by the Mr Spocks of this world. We have included many items of general information which, although strictly not part of the formal protocol description, provide a useful background on how APRS is actually used on the air, and how it is implemented in APRS software. We hope this will put APRS into perspective, will make the document more readable, and will not offend the purists too much.

It is important to realize how APRS originated, and to understand the design philosophy behind it. In particular, we feel strongly that APRS is, and should remain, a light-weight tactical system — almost anyone should be able to use it in temporary situations (such as emergencies or mobile work or weather watching) with the minimum of training and equipment.

This document is the result of inputs from many people, and collated and massaged by the APRS Working Group. Our sincere thanks go to everyone who has contributed in putting it together and getting it onto the street. If you discover any errors or omissions or misleading statements, please let us know — the best way to do this is via the TAPR aprsspec mailing list at www.tapr.org.

Finally, users throughout the world are continually coming up with new ideas and suggestions for extending and improving APRS. We welcome them. Again, the best way to discuss these is via the aprsspec list.

-The APRS Working Group, August 2000

Disclaimer

Like any navigation system, APRS is not infallible. No one should rely blindly on APRS for navigation, or in life-and-death situations. Similarly, this specification is not infallible.

¹Note: intentionally omitting some highlights from original

The members of the APRS Working Group have done their best to define the APRS protocol, but this protocol description may contain errors, or there may be omissions. It is very likely that not all APRS implementations will fully or correctly implement this specification, either today or in the future.

We urge anyone using or writing a program that implements this specification to exercise caution and good judgement. The APRS Working Group and the specification's Editor disclaim all liability for injury to persons or property that may result from the use of this specification or software implementing it.

Part I

The Structure of this Specification

This specification describes the overall requirements for developing software that complies with APRS Protocol Version 1.0. The information flow starts with the standard AX.25 UI-frame, and progresses downwards into more and more detail as the use of each field in the frame is explored. A key feature of the specification is the inclusion of dozens of detailed examples of typical APRS packets and related math computations. Here is an outline of the chapters:

Introduction to APRS - A brief background to APRS and a summary of its main features.

The APRS Design Philosophy - The fundamentals of APRS, highlighting its use as a real-time tactical communications tool, the timing of APRS transmissions and the use of generic digipeating.

APRS and AX.25 - A brief refresher on the structure of the AX.25 UI-frame, with particular reference to the special ways in which APRS uses the Destination and Source Address fields and the Information field.

APRS Data in the AX.25 Destination and Source Address Fields - Details of generic APRS callsigns and callsigns that specify display symbols and APRS software version numbers. Also a summary of how Mic-E encoded data is stored in the Destination Address field, and how the Source Address SSID can specify a display icon.

APRS Data in the AX.25 Information Field - Details of the principal constituents of APRS data that are stored in the Information field. Contains the APRS Data Type Identifiers table, and a summary of all the different types of data that the Information field can hold.

Time and Position Formats - Information on formats for timestamps, latitude, longitude, position ambiguity, Maidenhead locators, NMEA data and altitude.

APRS Data Extensions - Details of optional data extensions for station course/speed, wind speed/direction, power/height/gain, pre-calculated radio range, DF signal strength and Area Object descriptor.

Position and DF Report Data Formats - Full details of these report formats.

Compressed Position Report Data Formats - Full details of how station position and APRS data extensions are compressed into very short packets.

Mic-E Data Format - Mic-E encoding of station lat/long position, altitude, course, speed, Mic-E message code, telemetry data and APRS digipeater path into the AX.25 Destination Address and Information fields.

Object and Item Reports - Full information on how to set up APRS Objects and Items, and details of the encoding of Area Objects (circles, lines, ellipses etc).

Weather Reports - Full format details for weather reports from standalone (positionless) weather stations and for reports containing position information. Also details of storm data format.

Telemetry Data - A description of the MIM/KPC-3+ telemetry data format, with supporting information on how to tailor the interpretation of the raw data to individual circumstances.

Messages, Bulletins and Announcements - Full format information.

Station Capabilities, Queries and Responses - Details of the ten different types of query and expected responses.

Status Reports - The format of general status messages, plus the special cases of using a status report to contain meteor scatter beam heading/power and Maidenhead locator.

Network Tunneling - The use of the Source Path Header to allow tunneling of APRS packets through third-party networks that do not understand AX.25 addresses, and the use of the third-party Data Type Identifier.

User-Defined Data Format - APRS allows users to define their own data formats for special purposes. This chapter describes how to do this.

Other Packets - A general statement on how APRS is to handle any other packet types that are not covered by this specification.

APRS Symbols - How to specify APRS symbols and symbol overlays, in position reports and in generic GPS destination callsigns.

APRS Data Formats - An appendix containing all the APRS data formats collected together for easy reference.

The APRS Symbol Tables - A complete listing of all the symbols in the Primary and Alternate Symbol Tables.

ASCII Code Table - The full ASCII code, including decimal and hex codes for each character (the decimal code is needed for compressed lat/long and altitude computations), together with the hex codes for bit-shifted ASCII characters in AX.25 addresses (useful for Mic-E decoding and general on-air packet monitoring).

Glossary - A handy one-stop reference for the many APRS-specific terms used in this specification.

References - Pointers to other documents that are relevant to this specification.

Chapter 1

Chapter 1: Introduction to APRS

1.1 What is APRS?

APRS is short for Automatic Position Reporting System, which was designed by Bob Bruninga, WB4APR, and introduced by him at the 1992 TAPR/ ARRL Digital Communications Conference.

Fundamentally, APRS is a packet communications protocol for disseminating live data to everyone on a network in real time. Its most visual feature is the combination of packet radio with the Global Positioning System (GPS) satellite network, enabling radio amateurs to automatically display the positions of radio stations and other objects on maps on a PC. Other features not directly related to position reporting are supported, such as weather station reporting, direction finding and messaging.

APRS is different from regular packet in several ways:

- It provides maps and other data displays, for vehicle/personnel location and weather reporting in real time.
- It performs all communications using a one-to-many protocol, so that everyone is updated immediately.
- It uses generic digipeating, with well-known callsign aliases, so that prior knowledge of network topology is not required.
- It supports intelligent digipeating, with callsign substitution to reduce network flooding.
- Using AX.25 UI-frames, it supports two-way messaging and distribution of bulletins and announcements, leading to fast dissemination of text information.
- It supports communications with the Kenwood TH-D7 and TM-D700 radios, which have built-in TNC and APRS firmware.

Conventional packet radio is really only useful for passing bulk message traffic from point to point, and has traditionally been difficult to apply to real-time events where information has a very short lifetime. APRS turns packet radio into a real-time tactical communications and display system for emergencies and public service applications.

APRS provides universal connectivity to all stations, but avoids the complexity, time delays and limitations of a connected network. It permits any number of stations to exchange data just like voice users would on a voice net. Any station that has information to contribute simply sends it, and all stations receive it and log it.

APRS recognizes that one of the greatest real-time needs at any special event or emergency is the tracking of key assets. Where is the marathon leader? Where are the emergency vehicles? What's the weather at various points in the county? Where are the power lines down? Where is the head of the parade? Where is the mobile ATV camera? Where is the storm? To address these questions, APRS provides a fully featured automatic vehicle location and status reporting system. It can be used

over any two-way radio system including amateur radio, marine band, and cellular phone. There is even an international live APRS tracking network on the Internet.

1.2 APRS Features

APRS runs on most platforms, including DOS, Windows 3.x, Windows 95/98, MacOS, Linux and Palm. Most implementations on these platforms support the main features of APRS:

- **Maps** -APRS station positions can be plotted in real-time on maps, with coverage from a few hundred yards to worldwide. Stations reporting a course and speed are dead-reckoned to their present position. Overlay databases of the locations of APRS digipeaters, US National Weather Service sites and even amateur radio stores are available. It is possible to zoom in to any point on the globe.
- **Weather Station Reporting** - APRS supports the automatic display of remote weather station information on the screen.
- **DX Cluster Reporting** - APRS an ideal tool for the DX cluster user. Small numbers of APRS stations connected to DX clusters can relay DX station information to many other stations in the local area, reducing overall packet load on the clusters.
- **Internet Access** - The Internet can be used transparently to cross-link local radio nets anywhere on the globe. It is possible to telnet into Internet APRS servers and see hundreds of stations from all over the world live. Everyone connected can feed their locally heard packets into the APRS server system and everyone everywhere can see them.
- **Messages** - Messages are two-way messages with acknowledgement. All incoming messages alert the user on arrival and are held on the message screen until killed.
- **Bulletins and Announcements** - Bulletins and announcements are addressed to everyone. Bulletins are sent a few times an hour for a few hours, and announcements less frequently but possibly over a few days.
- **Fixed Station Tracking** - In addition to automatically tracking mobile GPS/LORAN-equipped stations, APRS also tracks from manual reports or grid squares.
- **Objects** - Any user can place an APRS Object on his own map, and within seconds that object appears on all other station displays. This is particularly useful for tracking assets or people that are not equipped with trackers. Only one packet operator needs to know where things are (e.g. by monitoring voice traffic), and as he maintains the positions and movements of assets on his screen, all other stations running APRS will display the same information.

Chapter 2

Chapter 2: APRS Design Philosophy

2.1 Net Cycle Time

It is important to note that APRS is primarily a real-time, tactical communications tool, to help the flow of information for things like special events, emergencies, Skywarn, the Emergency Operations Center and just plain in-the-field use under stress. But like the real world, for 99% of the time it is operating routinely, waiting for the unlikely serious event to happen.

Anything which is done to enhance APRS must not undermine its ability to operate in local areas under stress. Here are the details of that philosophy:

1. APRS uses the concept of a “net cycle time”. This is the time within which a user should be able to hear (at least once) all APRS stations within range, to obtain a more or less complete picture of APRS activity. The net cycle time will vary according to local conditions and with the number of digipeaters through which APRS data travels.
2. The objective is to have a net cycle time of 10 minutes for local use. This means that within 10 minutes of arrival on the scene, it is possible to capture the entire tactical picture.
3. All stations, even fixed stations, should beacon their position at the net cycle time rate. In a stress situation, stations are coming and going all the time. The position reports show not only where stations are without asking, but also that they are still active.
4. It is not reasonable to assume that all APRS users responding to a stress event understand the ramifications of APRS and the statistics of the channel — user settings cannot be relied on to avoid killing a stressed net. Thus, to try to anticipate when the channel is under stress, APRS automatically adjusts its net cycle time according to the number of digipeaters in the UNPROTO path:
 - Direct operation (no digipeaters): 10 minutes (probably an event).
 - Via one digipeater hop: 10 minutes (probably an event).
 - Via two digipeater hops: 20 minutes.
 - Via three or more digipeater hops: 30 minutes.
5. Since almost all home stations set their paths to three or more digipeaters, the default net cycle time for routine daily operation is 30 minutes. This should be a universal standard that everyone can bank on – if you routinely turn on your radio and APRS and do nothing else, then in 30 minutes you should have virtually the total picture of all APRS stations within range.
6. Since knowing where the digipeaters are located is fundamental to APRS connectivity, digipeaters should use multiple beacon commands to transmit position reports at different rates over different paths; i.e. every 10 minutes for sending position reports locally, and every 30 minutes for sending them via three digipeaters (plus others rates and distances as needed).

7. If the net cycle time is too long, users will be tempted to send queries for APRS stations. This will increase the traffic on the channel unnecessarily. Thus the recommended extremes for net cycle time are 10 and 30 minutes — this gives network designers the fundamental assumptions for channel loading necessary for good engineering design.

2.2 Packet Timing

Since APRS packets are error-free, but are not guaranteed delivery, APRS transmits information redundantly. To assure rapid delivery of new or changing data, and to preserve channel capacity by reducing interference from old data, APRS should transmit new information more frequently than old information.

There are several algorithms in use to achieve this:

- **Decay Algorithm** - Transmit a new packet once and n seconds later. Double the value of n for each new transmission. When n reaches the net cycle time, continue at that rate. Other factors besides “doubling” may be appropriate, such as for new message lines.
- **Fixed Rate** - Transmit a new packet once and n seconds later. Transmit it x times and stop.
- **Message-on-Heard** - Transmit a new packet according to either algorithm above. If the packet is still valid, and has not been acknowledged, and the net cycle time has been reached, then the recipient is probably not available. However, if a packet is then subsequently heard from the recipient, try once again to transmit the packet.
- **Time-Out** - This term is used to describe a time period beyond which it is reasonable to assume that a station no longer exists or is off the air if no packets have been heard from it. A period of 2 hours is suggested as the nominal default timeout. This time-out is not used in any transmitting algorithms, but is useful in some programs to decide when to cease displaying stations as “active”. Note that on HF, signals come and go, so decisions about activity may need to be more flexible.

2.3 Generic Digipeating

The power of APRS in the field derives from the use of generic digipeating, in that packets are propagated without a priori knowledge of the network. There are six powerful techniques which have evolved since APRS was introduced in 1992:

1. **RELAY** - Every VHF APRS TNC is assumed to have an alias of RELAY, so that anyone can use it as a digipeater at any time.
2. **ECHO** - HF stations use the alias of ECHO as an alternative to RELAY. (However, bearing in mind the nature of HF propagation, this has the potential of causing interference over a wide area, and should only be used sparingly by mobile stations).
3. **WIDE** - Every high-site digipeater is assumed to have an alias of WIDE for longer distance communications.
4. **TRACE** - Every high-site digipeater that is using callsign substitution is assumed to have the alias of TRACE. These digipeaters self-identify packets they digipeat by inserting their own call in place of RELAY, WIDE or TRACE.
5. **WIDEn-N** - A digipeater that supports WIDEn-N digipeating will digipeat any WIDEn-N packet that is “new” and will subtract 1 from the SSID until the SSID reaches -0. The digipeater keeps a copy or a checksum of the packet and will not digipeat that packet again within (typically) 28 seconds. This considerably reduces the number of superfluous digipeats in areas with many digipeaters in radio range of each other.
6. **GATE** - This generic callsign is used by HF-to-VHF Gateway digipeaters. Any packet heard on HF via GATE will be digipeated locally on VHF. This permits local networks to keep an eye on the national and international picture.

2.4 Communicating Map Views Unambiguously

APRS is a tactical geographical system. To maximize its operational effectiveness and minimize confusion between operators of different systems, users need to have an unambiguous way to communicate to others the “location” and “size” (or area of coverage) of any map view. The APRS convention is by reference to a center and range which specify the geographical center and approximate radius of a circle that will fit in the map view independent of aspect ratio. The radius of the circle (in nautical miles, statute miles or km) is known as the “range scale”. This convention gives all users a simple common basis for describing any specific map view to others over any communications medium or program.

Chapter 3

Chapter 3: APRS and AX.25

3.1 Protocols

At the link level, APRS uses the AX.25 protocol, as defined in Amateur Packet-Radio Link-Layer Protocol (see Appendix 6 for details), utilizing Unnumbered Information (UI) frames exclusively. This means that APRS runs in connectionless mode, whereby AX.25 frames are transmitted without expecting any response, and reception at the other end is not guaranteed. At a higher level, APRS supports a messaging protocol that allows users to send short messages (one line of text) to nominated stations, and expects to receive acknowledgements from those stations.

3.1.1 The AX.25 Frame

All APRS transmissions use AX.25 UI-frames, with 9 fields of data:

AX.25 UI-FRAME FORMAT									
FIELD NAME	Flag	Destination Address	Source Address	Digipeater Addresses (0-8)	Control Field (UI)	Protocol ID	Information Field	FCS	Flag
BYTES	1	7	7	0-56	1	1	1-256	2	1
FIELD NUMBER	One	Two	Three	Four	Five	Six	Seven	Eight	Nine

- **Flag-** The flag field at each end of the frame is the bit sequence 0x7e that separates each frame.
- **Destination Address-** This field can contain an APRS destination callsign or APRS data. APRS data is encoded to ensure that the field conforms to the standard AX.25 callsign format (i.e. 6 alphanumeric characters plus SSID). If the SSID is non-zero, it specifies a generic APRS digipeater path.
- **Source Address-** This field contains the callsign and SSID of the transmitting station. In some cases, if the SSID is non-zero, the SSID may specify an APRS display Symbol Code.
- **Digipeater Addresses-** From zero to 8 digipeater callsigns may be included in this field. Note: These digipeater addresses may be overridden by a generic APRS digipeater path (specified in the Destination Address SSID).
- **Control Field-** This field is set to 0x03 (UI-frame).
- **Protocol ID-** This field is set to 0xf0 (no layer 3 protocol).

- **Information Field-** This field contains more APRS data. The first character of this field is the APRS Data Type Identifier that specifies the nature of the data that follows.
- **Frame Check Sequence-** The FCS is a sequence of 16 bits used for checking the integrity of a received frame.
- **Flag-** The flag field at each end of the frame is the bit sequence 0x7e that separates each frame.

Chapter 4

Chapter 4: APRS Data in the AX.25 Destination and Source Address Fields

4.1 The AX.25 Destination Address Field

The AX.25 Destination Address field can contain 6 different types of APRS information:

- A generic APRS address.
- A generic APRS address with a symbol.
- An APRS software version number.
- Mic-E encoded data.
- A Maidenhead Grid Locator (obsolete).
- An Alternate Net (ALTNET) address.

In all of these cases, the Destination Address SSID may specify a generic APRS digipeater path.

4.2 Generic APRS Destination Addresses

APRS uses the following generic beacon-style destination addresses:

AIR* †	ALL*	AP*	BEACON	CQ*	GPS*	DF*
DGPS*	DRILL*	DX*	ID*	JAVA*	MAIL*	MICE*
QST*	QTH*	RTCM*	SKY*	SPACE*	SPC*	SYM*
TEL*	TEST*	TLM*	WX*	ZIP* †		

The asterisk is a wildcard, allowing the address to be extended (up to a total of 6 alphanumeric characters). Thus, for example, WX1, WX12 and WX12CD are all valid APRS destination addresses.

† The AIR* and ZIP* addresses are being phased out, but are needed at present for backward compatibility.

All of these addresses have an SSID of -0. Non-zero SSIDs are reserved for generic APRS digipeating. These addresses are copied by everyone. All APRS software must accept packets with these destination addresses.

The address GPS (i.e. the 3-letter address GPS, not GPS*) is specifically intended for use by trackers sending lat/long positions via digipeaters which have the capability of converting positions to compressed data format.

The addresses DGPS and RTCM are used by differential GPS correction stations. Most software will not make use of packets using this address, other than to pass them on to an attached GPS unit.

The address SKY is used for Skywarn stations.

Packets addressed to SPCL are intended for special events. APRS software can display such packets to the exclusion of all others, to minimize clutter on the screen from other stations not involved in the special event.

The addresses TEL and TLM are used for telemetry stations.

4.3 Generic APRS Address with Symbol

APRS uses several of the above-listed generic addresses in a special way, to specify not only an address but also a display symbol. These special addresses are GPSxyz, GPSCnn, GPSEnn, SPCxyz and SYMxyz, and are intended for use where it is not possible to include the symbol in the AX.25 Information field.

The GPS addresses above are for general use.

The SPC addresses are intended for special events.

The SYM addresses are reserved for future use.

The characters xy and nn refer to entries in the APRS Symbol Tables. The character z specifies a symbol overlay. See Chapter 20: APRS Symbols and Appendix Two for more information.

4.4 APRS Software Version Number

The AX.25 Destination Address field can contain the version number of the APRS software that is running at the station. Knowledge of the version number can be useful when debugging.

The following software version types are reserved (xx and xxx indicate a version number):

- APCxxx APRS/CE, Windows CE
- APDxxx Linux aprsd server
- APExxx PIC-Encoder
- APIxxx Icom radios (future)
- APICxx ICQ messaging
- APKxxx Kenwood radios
- APMxxx MacAPRS
- APPxxx pocketAPRS
- APRxxx APRSdos
- APRS older versions of APRSdos
- APRSM older versions of MacAPRS
- APRSW older versions of WinAPRS
- APSxxx APRS+SA
- APWxxx WinAPRS
- APXxxx X-APRS

- APYxxx Yaesu radios (future)
- APZxxx Experimental

This table will be added to by the APRS Working Group. For example, a station using version 3.2.6 of MacAPRS could use the destination callsign APM326.

The Experimental destination is designated for temporary use only while a product is being developed, before a special APRS Software Version address is assigned to it.

4.5 Mic-E Encoded Data

Another alternative use of the AX.25 Destination Address field is to contain Mic-E encoded data. This data includes:

- The latitude of the station.
- A West/East Indicator and a Longitude Offset Indicator (used in longitude computations).
- A Message Code.
- The APRS digipeater path.

This data is used with associated data in the AX.25 Information field to provide a complete Position Report and other information about the station (see Chapter 10: Mic-E Data Format).

4.6 Maidenhead Grid Locator in Destination Address

The AX.25 Destination Address field may contain a 6-character Maidenhead Grid Locator. For example: IO91SX. This format is typically used by meteor scatter and satellite operators who need to keep packets as short as possible.

This format is now obsolete.

4.7 Alternate Nets

Any other destination address not included in the specific generic list or the other categories mentioned above may be used in Alternate Nets (ALTNETs) by groups of individuals for special purposes. Thus they can use the APRS infrastructure for a variety of experiments without cluttering up the maps and lists of other APRS stations. Only stations using the same ALTNET address should see their data.

4.8 Generic APRS Digipeater Path

The SSID in the Destination Address field of all packets is coded to specify the APRS digipeater path. If the Destination Address SSID is -0, the packet follows the standard AX.25 digipeater ("VIA") path contained in the Digipeater Addresses field of the AX.25 frame.

If the Destination Address SSID is non-zero, the packet follows one of 15 generic APRS digipeater paths.

The SSID field in the Destination Address (i.e. in the 7th address byte) is encoded as follows:

APRS Digipeater Paths in Destination Address SSID

SSID	Path
-0	Use VIA path
-1	WIDE1-1
-2	WIDE2-2
-3	WIDE3-3
-4	WIDE4-4
-5	WIDE5-5
-6	WIDE6-6
-7	WIDE7-7

SSID	Path
-8	North path
-9	South path
-10	East path
-11	West path
-12	North path + WIDE
-13	South path + WIDE
-14	East path + WIDE
-15	West path + WIDE

4.9 The AX.25 Source Address SSID to specify Symbols

The AX.25 Source Address field contains the callsign and SSID of the originating station. If the SSID is -0, APRS does not treat it in any special way.

If, however, the Source Address SSID is non-zero, APRS interprets it as a display icon. This is intended for use only with stand-alone trackers where there is no other method of specifying a display symbol or a destination address (e.g. MIM trackers or NMEA trackers).

For more information, see Chapter 20: APRS Symbols.

Chapter 5

Chapter 5: APRS Data in the AX.25 Information Field

5.1 Generic Data Format

In general, the AX.25 Information field can contain some or all of the following information:

- APRS Data Type Identifier
- APRS Data
- APRS Data Extension
- Comment

<i>Generic APRS Information Field</i>				
	Data Type ID	APRS Data	APRS Data Extension	Comment
Bytes:	1	n	7	n

5.2 APRS Data Type Identifier

Every APRS packet contains an APRS Data Type Identifier (DTI). This determines the format of the remainder of the data in the Information field, as follows:

Note: There is one exception to the requirement for the Data Type Identifier to be the first character in the Information field — this is the Position without Timestamp (indicated by the ! DTI). The ! character may occur anywhere up to and including the 40th character position in the Information field. This variability is required to support X1J TNC digipeaters which have a string of unmodifiable text at the beginning of the field.

Note: The Kenwood TM-D700 radio uses the ' DTI for current Mic-E data. The radio does not use the ' DTI.

Table 5.1: **APRS Data Type Identifiers**

<i>Ident</i>	<i>Data Type</i>
0x1c	Current Mic-E Data (Rev 0 beta)
0x2d	Old Mic-E Data (Rev 0 beta)
!	Position without timestamp (no APRS messaging), or Ultimeter 2000 WX Station
"	[Unused]
#	Peet Bros U-II Weather Station
\$	Raw GPS data or Ultimeter 2000
%	Agrelo DFJr / MicroFinder
&	[Reserved —Map Feature]
'	Old Mic-E Data (but <i>Current data for TM-D700</i>)
(	[Unused]
)	Item
*	Peet Bros U-II Weather Station
+	[Reserved —Shelter data with time]
,	Invalid data or test data
-	[Unused]
.	[Reserved —Space weather]
/	Position with timestamp (no APRS messaging)
0-9	[Do not use]
:	Message
;	Object

Table 5.2: **APRS Data Type Identifiers, Continued**

<i>Ident</i>	<i>Data Type</i>
<	Station Capabilities
=	Position without timestamp (with APRS messaging)
>	Status
?	Query
@	Position with timestamp (with APRS messaging)
A-S	<i>[Do not use]</i>
T	Telemetry data
U-Z	<i>[Do not use]</i>
[	Maidenhead grid locator beacon (obsolete)
\	<i>[Unused]</i>
]	<i>[Unused]</i>
^	<i>[Unused]</i>
_	Weather Report (without position)
`	Current Mic-E Data (not used in TM-D700)
a-z	<i>[Do not use]</i>
{	User-Defined APRS packet format
	<i>[Do not use — TNC stream switch character]</i>
}	Third-party traffic
~	<i>[Do not use — TNC stream switch character]</i>

5.3 APRS Data and Data Extension

There are 10 main types of APRS Data:

- Position
- Direction Finding
- Objects and Items
- Weather
- Telemetry
- Messages, Bulletins and Announcements
- Queries
- Responses
- Status
- Other

Some of this data may also have an APRS Data Extension that provides additional information.

The APRS Data and optional Data Extension follow the Data Type Identifier. The table on the next page shows a complete list of all the different possible types of APRS Data and APRS Data Extension.

Possible APRS Data and Extension		
	Possible APRS Data	Possible APRS Data Extension
Position	Time (DHM or HMS) Lat/long coordinates Compressed lat/long/course/speed/radio range/altitude Symbol Table ID and Symbol Code Mic-E longitude, speed and course, telemetry or status Raw GPS NMEA sentente Raw weather station data	Course and Speed Power, Effective Antenna Height /Gain/Directivity Pre-Calculated Radio Range Omni DF Signal Strength Storm Data (in Comment field)
Direction Finding	Time (DHM or HMS) Lat/long coordinates Compressed lat/long/course/speed/radio range/altitude Symbol Table ID and Symbol Code	Course and Speed Power, Effective Antenna Height /Gain/Directivity Pre-Calculated Radio Range Omni DF Signal Strength Bearing and Number/Range/Quality (in Comment field)
Objects and Items	Object Name Item name Time (DHM or HMS) Lat/long coordinates Compressed lat/long/course/speed/radio range/altitude Symbol Table ID and Symbol Code Raw weather station data	Course and Speed Power, Effective Antenna Height Pre-Calculated Radio Range Omni DF Signal Strength Area Object Storm Data (in Comment field)
Weather	Time (MDHM) Lat/long coordinates Compressed lat/long/course/speed/radio range/altitude Symbol Table ID and Symbol Code Raw weather station data	Wind Direction and Speed Storm Data (in Comment field)
Telemetry	Telemetry (non Mic-E)	
Messages, Bulletins and Announcements	Addressee Message Text Message Identifier Message Acknowledgement Bulletin ID, Announcement ID Group Bulletin ID	
Queries	Query Type Query Target Footprint Addressee (Directed Query)	
Responses	Position Object/Item Weather Status Message Digipeater Trace Stations Heard Heard Statistics Station Capabilities	Course and Speed Power, Effective Antenna Height Pre-Calculated Radio Range Omni DF Signal Strength Area Object Wind Direction and Speed
Status	Time (DHM zulu) Status text Meteor Scatter Beam Heading/Power Maidenhead Locator (Grid Square) Altitude (Mic-E) 33 E-mail message	
Other	Things that don't fit in any of the other categories	

5.4 Comment Field

In general, any APRS packet can contain a plain text comment (such as a beacon message) in the Information field, immediately following the APRS Data or APRS Data Extension.

There is no separator between the APRS data and the comment unless otherwise stated.

The comment may contain any printable ASCII characters (except | and , which are reserved for TNC channel switching).

The maximum length of the comment field depends on the report — details are included in the description of each report.

In special cases, the Comment field can also contain further APRS data:

- Altitude in comment text (see Chapter 6: Time and Position Formats), or in Mic-E status text (see Chapter 10: Mic-E Data Format).
- Maidenhead Locator (grid square), in a Mic-E status text field (see Chapter 10: Mic-E Data Format) or in a Status Report (see Chapter 16: Status Reports).
- Bearing and Number/Range/Quality parameters (/BRG/NRQ), in DF reports (see Chapter 7: APRS Data Extensions).
- Area Object Line Widths (see Chapter 11: Object and Item Reports).
- Signpost Objects (see Chapter 11: Object and Item Reports).
- Weather and Storm Data (see Chapter 12: Weather Reports).
- Beam Heading and Power, in Status Reports (see Chapter 16: Status Reports).

5.5 Base-91 Notation

Two APRS data formats use base-91 notation: lat/long coordinates in compressed format (see Chapter 9) and the altitude in Mic-E format (see Chapter 10).

Base-91 data is compressed into a short string of characters. All the characters are printable ASCII, with character codes in the range 33–124 decimal (i.e. ! through |).

To compute the base-91 ASCII character string for a given data value, the value is divided by progressively reducing powers of 91 until the remainder is less than 91. At each step, 33 is added to the modulus of the division process to obtain the corresponding ASCII character code.

For example, for a data value of 12345678:

12345678 / 91³ = modulus 16, remainder 288542

288542 / 91² = modulus 34, remainder 6988

6988 / 91¹ = modulus 76, remainder 72

The four ASCII character codes are thus 49 (i.e. 16+33), 67 (i.e. 34+33), 109 (i.e. 76+33) and 105 (i.e. 72+33), corresponding to the ASCII string 1Cmi.

5.6 APRS Data Units

For historical reasons there is some lack of consistency between units of data in APRS packets — some speeds are in knots, others in miles per hour; some altitudes are in feet, others in meters, and so on. It is emphasized that this specification describes the units of data as they are transmitted on-air. It is the responsibility of APRS applications to convert the on-air units to more suitable units if required.

The default GPS earth datum is World Geodetic System (WGS) 1984.

Chapter 6

Chapter 6: Time and Position Formats

6.1 Time Formats

APRS timestamps are expressed in three different ways:

- Day/Hours/Minutes format
- Hours/Minutes/Seconds format
- Month/Day/Hours/Minutes format

In all three formats, the 24-hour clock is used.

Day/Hours/Minutes (DHM) format is a fixed 7-character field, consisting of a 6-digit day/time group followed by a single time indicator character (z or /). The day/time group consists of a two-digit day-of-the-month (01-31) and a four-digit time in hours and minutes.

Times can be expressed in *zulu* (UTC/GMT) or local time. For example:

092345z is 2345 hours zulu time on the 9th day of the month.

092345/ is 2345 hours local time on the 9th day of the month.

It is recommended that future APRS implementations only transmit zulu format on the air.

Note: The time in Status Reports may only be in zulu format.

Hours/Minutes/Seconds (HMS) format is a fixed 7-character field, consisting of a 6-digit time in hours, minutes and seconds, followed by the h time-indicator character. For example:

234517h is 23 hours 45 minutes and 17 seconds zulu.

Note: This format may not be used in Status Reports.

Month/Day/Hours/Minutes (MDHM) format is a fixed 8-character field, consisting of the month (01-12) and day-of-the-month (01-31), followed by the time in hours and minutes zulu. For example:

10092345 is 23 hours 45 minutes zulu on October 9th.

This format is only used in reports from stand-alone “positionless” weather stations (i.e. reports that do not contain station position information).

6.2 Use of Timestamps

When a station transmits a report without a timestamp, an APRS receiving station can make an internal record of the time it was received, if required. This record is the receiving station’s notion

of the time the report was created.

On the other hand, when a station transmits a report with a timestamp, that timestamp represents the transmitting station's notion of the time the report was created.

In other words, reports sent without a timestamp can be regarded as real-time, "current" reports (and the receiving station has to record the time they were received), whereas reports sent with a timestamp may or may not be realtime, and may possibly be (very) "old".

Four APRS Data Type Identifiers specify whether or not a report contains a timestamp, depending on whether the station has APRS messaging capability or not:

		No APRS Messaging	With APRS Messaging
(Current/real-time)	Report without timestamp	!	=
(Old/non-real-time)	Report with timestamp	/	@

Stations without APRS messaging capability are typically stand-alone trackers or digipeaters. Stations reporting without a timestamp are generally (but not necessarily) fixed stations.

6.3 Latitude Format

Latitude is expressed as a fixed 8-character field, in degrees and decimal minutes (to two decimal places), followed by the letter N for north or S for south.

Latitude degrees are in the range 00 to 90. Latitude minutes are expressed as whole minutes and hundredths of a minute, separated by a decimal point.

For example:

4903.50N is 49 degrees 3 minutes 30 seconds north.

In generic format examples, the latitude is shown as the 8-character string ddmm.hhN (i.e. degrees, minutes and hundredths of a minute north).

6.4 Longitude Format

Longitude is expressed as a fixed 9-character field, in degrees and decimal minutes (to two decimal places), followed by the letter E for east or W for west.

Longitude degrees are in the range 000 to 180. Longitude minutes are expressed as whole minutes and hundredths of a minute, separated by a decimal point.

For example:

07201.75W is 72 degrees 1 minute 45 seconds west.

In generic format examples, the longitude is shown as the 9-character string dddmm.hhW (i.e. degrees, minutes and hundredths of a minute west).

6.5 Position Coordinates

Position coordinates are a combination of latitude and longitude, separated by a display Symbol Table Identifier, and followed by a Symbol Code. For example:

4903.50N/07201.75W-

The / character between latitude and longitude is the Symbol Table Identifier (in this case indicating use of the Primary Symbol Table), and the - character at the end is the Symbol Code from that table (in this case, indicating a "house" icon).

A description of display symbols is included in Chapter 20: APRS Symbols. The full Symbol Table listing is in Appendix 2.

6.6 Position Ambiguity

In some instances — for example, where the exact position is not known — the sending station may wish to reduce the number of digits of precision in the latitude and longitude. In this case, the mm and hh digits in the latitude may be progressively replaced by a `_`(space) character as the amount of imprecision increases. For example:

4903.5_N represents latitude to nearest 1/10th of a minute.

4903.__N represents latitude to nearest minute.

490._._N represents latitude to nearest 10 minutes.

49._._._N represents latitude to nearest degree.

The level of ambiguity specified in the latitude will automatically apply to the longitude as well — it is not necessary to include any `_`characters in the longitude.

For example, the coordinates:

4903.VVN/07201.75W- represent the position to the nearest minute. That is, the hundredths of minutes of latitude and longitude may take any value in the range 00–99.

Thus the station may be located anywhere inside a bounding box having the following corner coordinates:

- North West corner: 49 deg 3.99 mins N, 72 deg 1.99 mins W
- North East corner: 49 deg 3.99 mins N, 72 deg 1.00 mins W
- South East corner: 49 deg 3.00 mins N, 72 deg 1.00 mins W
- South West corner: 49 deg 3.00 mins N, 72 deg 1.99 mins W

6.7 Default Null Position

Where a station does not have any specific position information to transmit (for example, a Mic-E unit without a GPS receiver connected to it), the station must transmit a default null position in the location field.

The null position corresponds to 0° 0' 0" north, 0° 0' 0" west. The null position should be (*sic*) include the symbol (unknown/indeterminate position).

For example, a Position Report for a station with unknown position will contain the coordinates:

...0000.00N\00000.00W...

6.8 Maidenhead Locator (Grid Square)

An alternative method of expressing a station's location is to provide a Maidenhead locator (grid square). There are four ways of doing this:

- In a Status Report — e.g. IO91SX/(- represents the symbol for a "house").
- In Mic-E Status Text — e.g. IO91SX/G (/G indicates a "grid square").
- In the Destination Address — e.g. IO91SX. (obsolete).
- In AX.25 beacon text, with the [APRS Data Type Identifier — e.g. [IO91SX] (obsolete).

Grid squares may be in 6-character form (as above) or in the shortened 4-character form (e.g. IO91).

6.9 NMEA Data

APRS recognizes raw ASCII data strings conforming to the NMEA 0183 Version 2.0 specification, originating from navigation equipment such as GPS and LORAN receivers. It is recommended that APRS stations interpret at least the following NMEA Received Sentence types:

- GGA Global Positioning System Fix Data
- GLL Geographic Position, Latitude/Longitude Data
- RMC Recommended Minimum Specific GPS/Transit Data
- VTG Velocity and Track Data
- WPT Way Point Location

6.10 Altitude

Altitude may be expressed in two ways:

- In the comment text.
- In Mic-E format.

Altitude in Comment Text - The comment may contain an altitude value, in the form /A=aaaaaa, where aaaaaa is the altitude in feet. For example: /A=001234. The altitude may appear anywhere in the comment.

Altitude in Mic-E format - The optional Mic-E status field can contain altitude data. See Chapter 10: Mic-E Data Format.

Chapter 7

Chapter 7: APRS Data Extensions

A fixed-length 7-byte field may follow APRS position data. This field is an APRS Data Extension. The extension may be one of the following:

7.1 Course and Speed

CSE/SPD	Course and Speed (this may be followed by a further 8 bytes containing DF bearing and Number/Range/Quality parameters)
DIR/SPD	Wind Direction and Wind Speed
PHGphgd	Station Power and Effective Antenna Height/Gain/ Directivity
RNGrrrr	Pre-Calculated Radio Range
DFSshgd	DF Signal Strength and Effective Antenna Height/Gain
Tyy/Cxx	Area Object Descriptor

The 7-byte CSE/SPD Data Extension can be used to represent the course and speed of a vehicle or APRS Object. The course is expressed in degrees (001-360), clockwise from due north. The speed is expressed in knots. A slash / character separates the two.

For example: 088/036 represents a course 88 degrees, traveling at 36 knots.

If the course and speed are unknown or not relevant, they can be set to 000/000 or .../... or ____/____.

Note: In the special case of DF reports, a course of 000 means that the DF station is fixed. If the course is non-zero, the station is mobile.

Table 7.1: PHG Codes

phgd Code:	0	1	2	3	4	5	6	7	8	9	Units
Power	0	1	4	9	16	25	36	49	64	81	watts
Height	10	20	40	80	160	320	640	1280	2560	5120	feet
Gain	0	1	2	3	4	5	6	7	8	9	dB
Directivity	omni	45 NE	90 E	135 SE	180 S	225 SW	270 W	315 NW	360 N		deg

7.2 Wind Direction and Wind Speed

The 7-byte DIR/SPD Data Extension can be used to represent the wind direction and sustained one-minute wind speed in a Weather Report. The wind direction is expressed in degrees (001-360), clockwise from due north. The speed is expressed in knots. A slash / character separates the two.

For example:

220/004 represents a wind direction of 220 degrees and a speed of 4 knots.

If the wind direction and speed are unknown or not relevant, they can be set to 000/000 or .../... or `000/000`.

7.3 Power, Effective Antenna Height/Gain/Directivity

The 7-byte PHGphgd Data Extension specifies the transmitter power, effective antenna height-above-average-terrain, antenna gain and antenna directivity. APRS uses this information to plot radio range circles around stations.

The 7 characters of this Data Extension are encoded as follows:

Characters 1-3:	PHG	(fixed)
Character 4:	p	Power code
Character 5:	h	Height code
Character 6:	g	Antenna gain code
Character 7:	d	Directivity code

The PHG codes are listed in the table below:

The height code represents the effective height of the antenna above average local terrain, not above ground or sea level — this is to provide a rough indication of the antenna's effectiveness in the local area.

The height code may in fact be any ASCII character 0-9 and above. This is so that larger heights for balloons, aircraft or satellites may be specified.

For example:

- : is the height code for 10240 feet (approximately 1.9 miles).
- ; is the height code for 20480 feet (approximately 3.9 miles), and so on.

The Directivity code offsets the PHG circle by one third in the indicated direction. This means a front-to-back ratio of 2 to 1. Most often this is used to indicate a favored direction or a null, even if an omni antenna is at the site.

Table 7.2: shgd Code:

shgd Code:	0	1	2	3	4	5	6	7	8	9	Units
Strength	0	1	2	3	4	5	6	7	8	9	S-points
Height	10	20	40	80	160	320	640	1280	2560	5120	feet
Gain	0	1	2	3	4	5	6	7	8	9	dB
Directivity	omni	45 NE	90 E	135 SE	180 S	225 SW	270 W	315 NW	360 N		deg

An example of the PHG Data Extension:

- PHG5132
- a power of 25 watts,
- an antenna height of 20 feet above the average local terrain,
- an antenna gain of 3 dB,
- and maximum gain due east.

7.4 Range Circle Plot

On receipt, APRS uses the p, h, g and d codes to calculate the usable radio range (in miles), for plotting a range circle representing the local radio horizon around the station. The radio range is calculated as follows: $\text{power} = p^2$

$\text{Height-above-average-terrain (haat)} = 10 \times 2h \text{ gain} = 10(g/10)$ $\text{range} = -(2 \times \text{haat} \times ((\text{power}/10) \times (\text{gain}/2)))$

Thus, for PHG5132: $\text{power} = 52 = 25 \text{ watts}$ $\text{haat} = 10 \times 21 = 20 \text{ feet}$ $\text{gain} = 10(3/10) = 1.995262$ $\text{range} = -(2 \times 20 \times ((25/10) \times (1.995262/2)))$

7.9 miles As the direction of maximum gain is due east, APRS will draw a range circle of radius 8 miles around the station, offset by 2.7 miles (i.e. one third of 8 miles) in an easterly direction. Note: In the absence of any PHG data, stations are assumed to be running 10 watts to a 3dB omni antenna at 20 feet, resulting in a 6-mile radius range circle, centered on the station.

7.5 Pre-Calculated Radio Range

The 7-byte RNgrrrr Data Extension allows users to transmit a precalculated omni-directional radio range, where rrrr is the range in miles (with leading zeros). For example, RNg0050 indicates a radio range of 50 miles. APRS can use this value to plot a range circle around the station.

7.6 Omni-DF Signal Strength

The 7-byte DFSshgd Data Extension lets APRS localize jammers by plotting the overlapping signal strength contours of all stations hearing the signal. This Omni-DF format replaces the PHG format to indicate DF signal strength, in that the transmitter power field is replaced with the relative signal strength (s) from 0 to 9.

7.6.1 DFS Codes

For example, DFS2360 represents a weak signal (around strength S2) heard on an omni antenna with 6 dB gain at 80 feet.

A signal strength of zero (0) is particularly significant, because APRS uses these 0 signal reports to draw (usually black) circles where the jammer is not heard. These black circles are extremely valuable since there will be a lot more reports from stations that do not hear the jammer than from those that do. This quickly eliminates a lot of territory.

7.7 Bearing and Number/Range/Quality

DF reports contain an 8-byte field /BRG/NRQ that follows the CSE/SPD Data Extension, specifying the course, speed, bearing and NRQ (Number/Range/ Quality) value of the report. NRQ indicates the Number of hits, the approximate Range and the Quality of the report.

For example, in: ...088/036/270/729...

course = 88 degrees, speed = 36 knots, bearing = 270 degrees, N = 7, R = 2, Q = 9

If N is 0, then the NRQ value is meaningless. Values of N from 1 to 8 give an indication of the number of hits per period relative to the length of the time period — thus a value of 8 means 100% of all samples possible got a hit. A value of 9 for N indicates to other users that the report is manual. The N value is not processed, but is just another indicator from the automatic DF units.

The range limits the length of the line to the original map's scale of the sending station. The range is 2R so, for R=4, the range will be 16 miles. Q is a single digit in the range 0-9, and provides an indication of bearing accuracy:

Q	Bearing Accuracy		Q	Bearing Accuracy
0	Useless		5	< 16 deg
1	< 240 deg		6	< 8 deg
2	< 120 deg		7	< 4 deg
3	< 64 deg		8	< 2 deg
4	< 32 deg		9	< 1 deg (best)

If the course and speed parameters are not appropriate, they should have the value 000/000 or .../... or `____/____`

7.8 Area Object Descriptor

The 7-byte Tyy/Cxx Data Extension is an Area Object Descriptor. The T parameter specifies the type of object (square, circle, triangle, etc) and the /C parameter specifies its fill color. Area Objects are described in Chapter 11: Object and Item Reports.

Chapter 8

Chapter 8: Position and DF Report Data Formats

8.1 Position Reports

Lat/Long Position Reports are contained in the Information field of an APRS AX.25 frame. The following diagrams show the permissible formats of these reports, together with some examples. The gray areas indicate optional fields, and the shaded (yellow) characters are literal ASCII characters. In all cases there is a maximum of 43 characters after the Symbol Code.

Lat/Long Position Report Format — without Timestamp						
	! or =	Lat	Sym Table ID	Long	Symbol Code	Comment (max 43 chars)
Bytes:	1	8	1	9	1	0-43

Examples	
!4903.50N/07201.75W-Test 001234 (no timestamp, no APRS messaging, with comment.)	
!4903.50N/07201.75W-Test /A=001234 (no timestamp, no APRS messaging, altitude = 1234 ft.)	
!49░░.N/072░░.░░W- (no timestamp, no APRS messaging, location to nearest degree.)	
TheNet░X-1J4░░(BFLD)!4903.50N/07201.75Wn (no timestamp, no APRS messaging...,)	
(... with X1J node header string.)	

Table 8.1: Lat/Long Position Report Format — without Timestamp

Lat/Long Position Report Format — with Timestamp

	/ or @	Time DHM / HMS	Lat	Sym Table ID	Long	Symbol Code	Comment (max 43 chars)
Bytes	1	7	8	1	9	1	0-43

Examples

/092345z4903.50N/07201.75W>Test1234 with timestamp
no APRS messaging, zulu time, with comment.
@092345/4903.50N/07201.75W>Test1234 with timestamp,
with APRS messaging, local time, with comment.

Table 8.2: Lat/Long Position Report Format — with Timestamp

Lat/Long Position Report Format — with Data Extension (no Timestamp)

	! or =	Lat	Sym Table	Long	Symbol Code	Data Ext †	Comment (max 36 chars)
Bytes:	1	8	1	9	1	7	0-36

Where † is one of:

- Course/Speed
- Power/Height/Gain/Dir
- Radio Range
- DF Signal Strength

Examples

=4903.50N/07201.75W#PHG5132 no timestamp, with APRS messaging, with PHG.
=4903.50N/07201.75W_225/000g000t050r000p001...h00b10138dU2k weather report.

Table 8.3: Lat/Long Position Report Format — with Data Extension (no Timestamp)

Lat/Long Position Report Format — with Data Extension and Timestamp

	/ or @	Time DHM / HMS	Lat	Sym Table	Long	Symbol Code	Data Ext †	Comment (max 36 chars)
Bytes:	1	7	8	1	9	1	7	0-36

Where † is one of:

- Course/Speed
- Power/Height/Gain/Dir
- Radio Range
- DF Signal Strength

Examples

- @092345/4903.50N/07201.75W>088/036
(with timestamp, with APRS messaging, local time,course/speed.)
- @234517h4903.50N/07201.75W>PHG5132
(with timestamp, APRS messaging, hours/mins/secs time, PHG.)
- @092345z4903.50N/07201.75W>RNG0050
(with timestamp, APRS messaging, zulu time, radio range.)
- /234517h4903.50N/07201.75W>DFS2360
(with timestamp, hours/mins/secs time, DF,no APRS messaging.)
- @092345z4903.50N/07201.75W_090/000g000t066r000p000...dUII
(weather report.)

Table 8.4: Lat/Long Position Report Format — with Data Extension and Timestamp

Maidenhead Locator Beacon				
	[	Grid Locator	]	Comment
Bytes	1	4 or 6	1	n

Examples

[IO91SX] 35 miles NNW of London
[IO91]

Table 8.5: Maidenhead Locator Beacon

Raw NMEA Position Report Format		
NMEA Received Sentence		
	\$	
Bytes	1	25-209

Examples

```
$GPGGA,102705,5157.9762,N,00029.3256,W,1,04,2.0,75.7,M,47.6,M,,*62
$GPGLL,2554.459,N,08020.187,W,154027.281,A
$GPRMC,063909,A,3349.4302,N,11700.3721,W,43.022,89.3,291099,13.6,E*52
$GPVTG,318.7,T,,M,35.1,N,65.0,K*69
```

Table 8.6: Raw NMEA Position Report Format

8.2 DF Reports

DF Reports (Direction Finding)¹

DF Reports are contained in the Information field of an APRS AX.25 frame. The Bearing and Number/Range/Quality (BRG/NRQ) parameters follow the Data Extension field. Note: The BRG/NRQ parameters are only meaningful when the report contains the DF symbol (i.e. the Symbol Table ID is / and the Symbol Code is \).

Note: If the DF station is fixed, the Course value is zero. If the station is moving, the Course value is non-zero.

¹Added by Ed.

DF Report Format — without Timestamp

	! or =	Lat	Sym Table ID /	Long	Symbol Code \ /	Data Ext †	/BRG/NRQ	Comment (max 28 chars)
Bytes:	1	8	1	9	1	7	8	0-28

Where † is one of:

- Course/Speed
- Power/Height/Gain/Dir
- Radio Range
- DF Signal Strength

Examples

=4903.50N/07201.75W\088/036/270/729

=4903.50N/07201.75W\000/036/270/729

- First example is: no timestamp, course/speed/ bearing/NRQ, with APRS messaging. DF station moving (CSE is non-zero).
 - Second example is: Same report, DF station fixed (CSE=000).
-

Table 8.7: DF Report Format – without Timestamp

DF Report Format — with Timestamp

	/ or @	Time DHM/ HMS	Lat	Sym Table ID /	Long	Symbol Code \ /	Data Ext †	/BRG/NRQ	Comment (max 28 chars)
Bytes:	1	7	8	1	9	1	7	8	0-28

Where † is one of:

- Course/Speed
- Power/Height/Gain/Dir
- Radio Range
- DF Signal Strength

Examples

@092345z4903.50N/07201.75W\088/036/270/729
/092345z4903.50N/07201.75W\000/000/270/729

- First example is: with timestamp, course/speed/ bearing/NRQ, with APRS messaging.
 - Second example is: with timestamp, bearing/NRQ, no course/speed, no APRS messaging.
-

Table 8.8: DF Report Format with Timestamp

Needs work

Chapter 9

Chapter 9: Compressed Position Report Data Formats

In compressed data format, the Information field contains the station's latitude and longitude, together with course and speed or pre-calculated radio range or altitude.

This information is compressed to minimize the length of the transmitted packet (and therefore improve its chances of being received correctly under less than ideal conditions).

The Information field also contains a display Symbol Code, and there may optionally be a plain text comment (uncompressed) as well.

9.1 The Advantages of Data Compression

Compressed data format may be used in place of the numeric lat/long coordinates already described, such as in the !, /, @ and = formats. Data compression has several important benefits:

- Fully backwards compatible with all existing formats.
- Fully supports any comment string.
- Speed is accurate to +/-1 mph up to about 40 mph and within 3% at 600 mph.
- Altitude in feet is accurate to +/- 0.4% from 1 foot to 3000 miles.
- Consistent one-algorithm processing of compressed latitude and longitude.
- Improved position to 1 foot worldwide.
- Pre-calculated radio range, compressed to one byte.
- Potential 50% compression of every position format on the air.
- Potential 40% reduction of raw GPS NMEA data length.
- Additional 7-byte reduction for NEMA GGA altitudes.
- Support for TNC compression at the NMEA source (from the GPS receiver).
- Digipeater compression of old NMEA trackers on the fly.
- Usage is optional in all cases.

The only minor disadvantages are that the course only resolves to +/- 2 degrees, and this format does not support PHG.

9.2 Compressed Data Format

Compressed data may be generated in several ways:

- by APRS software.
- pre-entered manually into a digipeater's beacon text.
- by a digipeater converting raw tracker NMEA packets to compressed.

[In future, there is the possibility that a Kantronics KPC-3 or other tracker TNC will be able to compress data directly from an attached GPS receiver]. In all cases the compressed format is a fixed 13-character field:

/YYYYXXXX\$csT

where

- / is the Symbol Table Identifier
- YYYY is the compressed latitude
- XXXX is the compressed longitude
- \$ is the Symbol Code
- cs is
 - the compressed course/speed;
 - compressed pre-calculated radio range;
 - or compressed altitude
- T is the compression type indicator

Compressed Position Data					
Sym Table	Compressed Lat	Compressed Long	Symbol Code	Compressed Course/Speed	Comp
				Compressed Radio Range	Type
Id	YYYY	XXXX		Compressed Altitude	T
1	4	4	1	2	1

Compressed format can be used in place of lat/long position format anywhere that ...ddmm.hhN/dddmm.hhW\$xxxxxxx... occurs.

All bytes except for the / and \$ are base-91 printable ASCII characters (!..{). These are converted to numeric values by subtracting 33 from the decimal ASCII character code. For example, # has an ASCII code of 35, and represents a numeric value of 2 (i.e. 35-33).

9.3 Symbol

The presence of the leading Symbol Table Identifier instead of a digit indicates that this is a compressed Position Report and not a normal lat/long report.

9.4 Lat/Long Encoding

The values of YYYY and XXXX are computed as follows:

YYYY is $380926 \times (90 - \text{latitude})$ [base 91] latitude is positive for north, negative for south, in degrees.
XXXX is $190463 \times (180 + \text{longitude})$ [base 91] longitude is positive for east, negative for west, in degrees.

For example, for a longitude of $72^\circ 45' 00''$ west (i.e. -72.75 degrees), the math is $190463 \times (180 - 72.75) = 20427156$. Because this is to base 91, it is then necessary to progressively divide this value by reducing powers of 91, to obtain the numerical values of X:

$20427156 / 91^3 = 27$, remainder 80739
 $80739 / 91^2 = 9$, remainder 6210
 $6210 / 91^1 = 68$, remainder 22

To obtain the corresponding ASCII characters, 33 is added to each of these values, yielding 60 (i.e. $27+33$), 42, 101 and 55. From the ASCII Code Table (in Appendix 3), this corresponds to <*e7 for XXXX.

9.5 Lat/Long Decoding

To decode a compressed lat/long, the reverse process is needed. That is, if YYYY is represented as $y_1y_2y_3y_4$ and XXXX as $x_1x_2x_3x_4$, then:

$$\text{Lat} = 90 - ((y_1 - 33) \times 91 + (y_2 - 33) \times 91 + (y_3 - 33) \times 91 + y_4 - 33) / 380926$$

$$\text{Long} = -180 + ((x_1 - 33) \times 91 + (x_2 - 33) \times 91 + (x_3 - 33) \times 91 + x_4 - 33) / 190463$$

For example, if the compressed value of the longitude is <*e7 (as computed above), the calculation becomes:

$$\begin{aligned} \text{Long} &= -180 + (27 \times 91 + 9 \times 91 + 68 \times 91 + 22) / 190463 \\ &= -180 + (20346417 + 74529 + 6188 + 22) / 190463 \\ &= -180 + 107.25 \\ &= -72.75 \text{ degrees} \end{aligned}$$

9.6 Course/Speed, Pre-Calculated Radio Range and Altitude

The two cs bytes following the Symbol Code character can contain either the compressed course and speed or the compressed pre-calculated radio range or the station's altitude. These two bytes are in base 91 format.

In the special case of $c = _$ (space), there is no course, speed or range data, in which case the csT bytes are ignored.

Course/Speed — If the ASCII code for c is in the range ! to z inclusive — corresponding to numeric values in the range 0-89 decimal (i.e. after subtracting 33 from the ASCII code) — then cs represents a compressed course/speed value:

$$\text{course} = c \times 4$$

$$\text{speed} = 1.08^s - 1$$

For example, if the cs characters are 7P, the corresponding values of c and s (after subtracting 33 from the ASCII character code) are 22 and 47 respectively. Substituting these values in the above equations:

$$\text{course} = 22 \times 4 = 88 \text{ degrees}$$

$$\text{speed} = 1.0847 - 1 = 36.2 \text{ knots}$$

Pre-Calculated Radio Range — If $c = \{$, then cs represents a compressed pre-calculated radio range value:

$\text{range} = 2 \times 1.08^s$

For example, if the cs bytes are ?, the ASCII code for ? is 63, so the value of s is 30 (i.e. 63-33). Thus:

$\text{range} = 2 \times 1.08^{30} \approx 20 \text{ miles}$

So APRS will draw a circle of radius 20 miles around the station plot on the screen.

9.7 The Compression Type (T) Byte

The T byte follows the cs bytes. The T byte contains several bit fields showing the GPS fix status, the NMEA source of the position data and the origin of the compression.

The T byte is not meaningful if the c byte is ␣(space).

Compression Type (T) Byte Format								
Bit:	7	6	5	4	3	2	1	0

NMEA Source 0 0 = other 0 1 = GLL 1 0 = GGA 1 1 = RMC

Compression Origin 0 0 0 = Compressed 0 0 1 = TNC BText 0 1 0 = Software (DOS/Mac/Win/+SA)
0 1 1 = [tbd] 1 0 0 = KPC3 1 0 1 = Pico 1 1 0 = Other tracker [tbd] 1 1 1 = Digipeater conversion

For example, if the compressed position was derived from an RMC sentence, the fix is current, and the compression was performed by APRSdos software, then the value of T in binary is 0 0 1 11 010, which equates to 58 decimal. Adding 33 to this value gives the ASCII code for the T byte (i.e. 91), which corresponds to the [character.

Thus, using data from all the earlier examples, if the RMC sentence contains (among other parameters) the following data:

Latitude = 49° 30' 00" north

Longitude = 72° 45' 00" west

Speed = 36.2 knots

Course = 88°

and:

the fix is current

compression is performed by APRSdos software

the display symbol is a "car"

then the complete 13-character compressed location field is transmitted as:

/	YYYY	XXXX	\$	csT
/	5L!!	<*e7	>	7P[

9.8 Altitude

If the T byte indicates that the raw data originates from a GGA sentence (i.e. bits 4 and 3 of the T byte are 10), then the sentence contains an altitude value, in feet. After compression, the compressed altitude data is placed in the cs bytes, such that:

$\text{altitude} = 1.002\text{cs feet}$

For example, if the received cs bytes are S], the computation is as follows:

Subtract 33 from the ASCII code for each character:

$c = 83 - 33 = 50$

$s = 93 - 33 = 60$

Multiply c by 91 and add s to obtain cs : $cs = 50 \times 91 + 60$
 $= 4610$

Then altitude $= 1.002^{4610}$
 $= 10004$ feet

9.9 New Trackers

Tracker firmware may compress GPS data directly to APRS compressed format. They would use the ! Data Type Indicator, showing that the position is real-time and that the tracker is not APRS-capable. If the Position Report is not real-time, then the **!** Data Type Indicator can be used instead, so that the latest fix time may be included.

Some digipeaters have the ability to convert raw NMEA strings from existing trackers to compressed data format for further forwarding.

9.10 Old Trackers

These digipeaters will compress the data if the tracker Destination Address is GPS. (Note: This is the 3-letter address GPS, not GPS*). Trackers desiring for their packets to not be modified by the APRS network will use any other valid generic APRS Destination Address.

Compressed data is contained in the AX.25 Information field, in these formats:

9.11 Compressed Report Formats

Examples `=/5L!!<*e7>V sTComment`

2

with APRS messaging. Note the V space character following the > Symbol Code, indicating that there is no course/speed, radio range or altitude. The sT characters are fillers and have no significance here. with APRS messaging, RMC sentence, with course/speed. with APRS messaging, with radio range. with APRS messaging, GGA sentence, altitude.

`=/5L!!<*e7>7P[ =/5L!!<*e7>?! =/5L!!<*e7OS]S`

Compressed Lat/Long Position Report Format — with Timestamp

Example `@092345z/5L!!<*e7>?!`

with APRS messaging, timestamp, radio range.

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Chapter 12: Weather Reports

12.1 Weather Report Types

APRS is an ideal tool for reporting weather conditions via packet. APRS supports serial data transmissions from the Peet Brothers, Ultimeter and Davis home weather stations. It is even possible to mount an Ultimeter remotely with only a TNC and radio to report and plot conditions. APRS is also ideally suited for the Skywarn weather observer initiative. APRS supports three types of Weather Report:

12.2 Data Type Identifiers

- Raw Weather Report
- Positionless Weather Report
- Complete Weather Report

The following APRS Data Type Identifiers are used in Weather Reports containing raw data:

! Ultimeter 2000

Peet Bros U-II

\$ Ultimeter 2000

* Peet Bros U-II

Positionless weather data

In addition, where the raw data has been post-processed (for example, by the insertion of station location information), the four position Data Type Identifiers !, #, / and @ may be used instead. In this case, the Weather Report is identified with the weather symbol /_ or _ in the APRS Data.

12.3 Raw Weather Reports

Raw weather data from a stand-alone weather station is contained in the Information Field of an APRS AX.25 frame:

Raw Weather Report Format		
	! or # or \$ or *	Raw Weather Data
Bytes:	1	n

Examples

Ultimeter 2000: !!006B005803500000—03E9—002105140000005D (Ultimeter 2000)

Peet Bros U-II: #50B7500820082

Ultimeter 2000: \$ULTW0031003702CE0069—000086A00001—011901CC00000005

Peet Bros U-II: *7007600000000

12.4 Positionless Weather Reports

Generic raw weather data from a stand-alone weather station is contained in the Information Field of an APRS AX.25 frame:

Positionless Weather Report Format					
	—	Time (MDHM)	Positionless Weather Data	APRS Software (S)	WX Unit (uuuu)
Bytes:	1	8	n	1	2-4

Example

Report derived from Radio Shack WX station data:

_10090556c220s004g005t077r000p000P000h50b09900wRSW

12.4.1 APRS Software Type

A Weather Report may contain a single-character code S for the type of APRS software that is running at the weather station:

d = APRSdos

M = MacAPRS

P = pocketAPRS

S = APRS+SA

W = WinAPRS

X = X-APRS (Linux)

Weather Unit Type

A Weather Report may contain a 2-4 character code uuuu for the type of weather station unit. The following codes have been allocated:

Dvs = Davis

HKT = Heathkit

PIC = PIC device

RSW = Radio Shack

U-II = Original Ultimeter U-II (auto mode)

U2R = Original Ultimeter U-II (remote mode)

U2k = Ultimeter 500/2000

U2kr = Remote Ultimeter logger

U5 = Ultimeter 500

Upkm = Remote Ultimeter packet mode

Users may specify any other 2-4 character code for devices not in this list.

12.4.2 Positionless Weather Data

The format of weather data within a Positionless Weather Report differs according to the type of weather station unit, but generically consists of some or all of the following elements:

Positionless Weather Data									
	Wind Direction	Wind Speed	Gust	Temp (F)	Rain Last Hr	Last 24 Hrs	Rain Since Midnight	Humidity	Barometric Pressure
	cccc	ssss	gggg	tttt	rrrr	pppp	PPPP	hhh	bbbbbb
Bytes	4	4	4	4	4	4	4	3	5

Where:

- c = wind direction (in degrees).
- s = sustained one-minute wind speed (in mph).
- g = gust (peak wind speed in mph in the last 5 minutes).
- t = temperature (in degrees Fahrenheit). Temperatures below zero are expressed as -01 to -99.
- r = rainfall (in hundredths of an inch) in the last hour.
- p = rainfall (in hundredths of an inch) in the last 24 hours.
- P = rainfall (in hundredths of an inch) since midnight.
- h = humidity (in %. 00 = 100%).
- b = barometric pressure (in tenths of millibars/tenths of hPascal).

Other parameters that are available on some weather station units include:

- L = luminosity (in watts per square meter) 999 and below.
- l (lower-case letter "L") = luminosity (in watts per square meter) 1000 and above.

- NB: (L is inserted in place of one of the rain values).
- s= snowfall (in inches) in the last 24 hours.
- # = raw rain counter

Note: The weather report must include at least the MDHM date/timestamp, wind direction, wind speed, gust and temperature, but the remaining parameters may be in a different order (or may not even exist). Note: Where an item of weather data is unknown or irrelevant, its value may be expressed as a series of dots or spaces. For example, if there is no wind speed/direction/gust sensor, the wind values could be expressed as:

- c...s...g...
- or: c____s____g ____

For example, Jim's rain gauge may produce a report like this:
_10090556c...s...g...t...P012Jim

(The date/timestamp, wind direction/speed/gust and temperature parameters must be included, even though they are not meaningful).

12.4.3 Location of a Raw and Positionless Weather Stations

APRS cannot display weather data on a map until it knows the location of the sending station. In the case of a station transmitting Raw or Positionless Weather Reports, the station has to occasionally send an additional packet containing its position (using any of the legal lat/long and compressed lat/long position formats described earlier).

12.4.4 Symbols with Raw and Positionless Weather Stations

Because Raw and Positionless Weather Reports do not contain a display symbol in the AX.25 Information field, it is possible to specify the symbol in a generic APRS destination address (e.g. GPSTHW or GPSE63) instead. Alternatively, if the weather station is on a balloon, the SSID -11 may be used in the source address (e.g. N0QBF-11).

See Chapter 20: APRS Symbols for more detail on the usage of symbols.

12.5 Complete Weather Reports with Timestamp and Position

An APRS Complete Weather Report can contain a timestamp and location information, using any of the legal lat/long and compressed lat/long position formats described earlier. An APRS Object may also have weather information associated with it.

Examples of report formats are shown below. Note that the Symbol Code in every case is the _ (underscore). Also, the 7-byte Wind Direction and Wind Speed Data Extension replace the cccc and ssss fields of a Positionless Weather Report.

Complete Weather Report Format — with Lat/Long position, no Timestamp									
	! or =	Lat	Sym Table ID	Long	Symbol Code –	Wind Direction Speed	Weather Data	APRS Software S	WX Unit uuuu
Bytes	1	8	1	9	1	7	n	1	2-4

Examples

!4903.50N/07201.75W_220/004g005t077r000p000P000h50b09900wRSW

!4903.50N/07201.75W_220/004g005t077r000p000P000h50b.....wRSW

Complete Weather Report Format — with Lat/Long position and Timestamp										
	/ or @	Time DHM/ HMS	Lat	Sym Table ID	Long	Symbol Code –	Wind Direction Speed	Weather Data	APRS Software S	WX Unit uuuu
Bytes	1	7	8	1	9	1	7	n	1	2-4

Example

@092345z4903.50N/07201.75W_220/004g005t-07r000p000P000h50b09900wRSW

Complete Weather Report Format — with Compressed Lat/Long position, no Timestamp

	! or =	Sym Table ID	Comp Lat YYYY	Comp Long XXXX	Symbol Code _	Comp Wind Dir + Speed	Comp Type T	Weather Data	APRS Software S	WX Unit uuuu
Bytes	1	1	4	4	1	2	1	n	1	2-4

Example

=/5L!!<*e7>_7P[g005t077r000p000P000h50b09900wRSW

Complete Weather Report Format — with Compressed Lat/Long position, with Timestamp

	/ or @	Time DHM HMS	Sym Table ID	Comp Lat YYYY	Comp Long XXXX	Symbol Code _	Comp Wind Dir + Speed	Comp Type T	Weather Data	APRS Software S	WX Unit uuuu
Bytes											
	1	7	1	4	4	1	2	1	n	1	2-4

Example

@092345z/5L!!<*e7>_7P[g005t077r000p000P000h50b09900wRSW

Complete Weather Report Format — with Object and Lat/Long position

		Object Name	*	Time DHM/ HMS	Lat	Sym Table ID	Long	Symbol	Wind Directn/ Speed	Weather Data	APRS Software S	WX Unit uuuu
Bytes												
	1	9	1	7	8	1	9	1	7	n	1	2-4

Examples

;BRENDVVV*4903.50N/07201.75W_220/004g005t077r000p000P000h50b09900wRSW

;BRENDVVV*092345z4903.50N/07201.75W_220/004g005b0990

12.6 Storm Data

APRS reports can contain data relating to tropical storms, hurricanes and tropical depressions. The format of the data is as follows:

Storm Data										
	Direction	/	Speed	Storm Type	Sustained Wind Speed	Peak Wind Gusts	Central Pressure	Radius Hurricane Winds	Radius Tropical Storm	Radius Whole Gale
				/ST	/www	^GGG	/ppp	>RRR	&rrr	%ggg
Bytes	3	1	3	3	4	4	5	4	4	4

Where ST is one of:

= TS (Tropical Storm)
= HC (Hurricane)
= TD (Tropical Depression)

and

WWW = sustained wind speed (in knots).

GGG = gust (peak wind speed in knots).

pppp = central pressure (in millibars/hPascal)

RRR = radius of hurricane winds (in nautical miles).

rrr = radius of tropical storm winds (in nautical miles).

ggg = radius of “whole gale” (i.e. 50 knot) winds (in nautical miles). Optional.

Storm data will usually be included in an Object Report, but may also be included in a Position Report or an Item Report.

The display symbol will be either:

 Hurricane/Tropical Storm (current position)

 Hurricane (predicted future position)

For example, the progress of Hurricane Brenda could be expressed in Object Reports like these:

```
;BRENDVVV*092345z4903.50N\07202.75W@088/036/HC/150^200/0980>090&030%040  
;BRENDVVV*100045z4905.50N/07201.75W@101/047/HC/104^123/0980>065&020%040
```

12.7 National Weather Service Bulletins

APRS supports the dissemination of National Weather Service bulletins. See Chapter 14: Messages, Bulletins and Announcements.

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Chapter 21

APRS Symbols

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Part II

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Part III

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Part IV

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